

Roll No.

TOE-801

B. TECH. (EE) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April/May, 2022
EXPERT SYSTEM AND FUZZY LOGIC

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What is Fuzzy Logic ? Explain discrete and continuous fuzzy set with examples. Also, explain membership value. 10 Marks (CO4, CO3)

OR

- (b) What is crisp logic ? How do we represent a crisp set ? Explain the different crisp set operations with the help of examples.

10 Marks (CO3, CO4)

2. (a) Find : 10 Marks (CO2)

$$A = (1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 9)$$

$$B = (2 \ 4 \ 5 \ 7 \ 8)$$

(i) $A - B$

(ii) $A \cap B$

(iii) $A \cup B$

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(iv) $\overline{(A \cup B)}$

(v) $\overline{(A \cap B)}$

(vi) $A \cup (A \cap B)$

(vii) $A \cap (A \cup B)$

OR

(b) Explain the following properties of fuzzy set : 10 Marks (CO2, CO3)

(i) Law of excluded middle

(ii) Law of contradiction

(iii) Entropy of a fuzzy set

3. (a) Explain the various types of membership function used to represent fuzzy set. Also show the mathematical representation of these membership functions. 10 Marks (CO2, CO3)

OR

(b) The membership function distribution of a fuzzy set is assumed to follow a Gaussian distribution with mean, $m = 100$ and standard deviation of 20. Determine 0.6-cut of this distribution.

10 Marks (CO2, CO3)

4. (a) A fuzzy set $A(x)$ is given as follows : 10 Marks (CO2, CO3)

$$A(x) = \{(x_1, 0.1) (x_2, 0.2) (x_3, 0.3) (x_4, 0.4)\}$$

Find its scalar cardinality and height of $A(x)$.

OR

(b) Explain : 10 Marks (CO2, CO3)

(i) Complement of a fuzzy set

(3)

(ii) Intersection of a fuzzy set

(iii) Union of a fuzzy set

(iv) Power of a fuzzy set

5. (a) Find :

10 Marks (CO2, CO3)

$$A(x) = \{(x_1, 0.1) (x_2, 0.2) (x_3, 0.3) (x_4, 0.4)\}$$

$$B(x) = \{(x_1, 0.5) (x_2, 0.7) (x_3, 0.8) (x_4, 0.9)\}$$

(i) $A(x).B(x)$

(ii) $4.A(x)$

(iii) $A^2(x)$

(iv) $A(x) + B(x)$

(v) $A(x) \oplus B(x)$

(vi) $A(x) - B(x)$

OR

(b) A and B are two fuzzy relations expressed in matrix form. Find the composition of these two fuzzy relations, that is matrix C.

$$A = \begin{bmatrix} 0.2 & 0.3 \\ 0.5 & 0.7 \end{bmatrix}$$

$$B = \begin{bmatrix} 0.3 & 0.6 & 0.7 \\ 0.1 & 0.8 & 0.6 \end{bmatrix}$$

10 Marks (CO2, CO3)

Roll No.

TEE-802

B. TECH. (EE) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April/May, 2022
WIND AND SOLAR ENERGY SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What factors led to accelerated development of wind power ?

10 Marks (CO1, CO2)

OR

- (b) Derive an expression for energy available in wind. What are the factors responsible for distribution of wind energy on the surface of the earth ?

10 Marks (CO1, CO2)

2. (a) Comment on the relative features of HAWT and VAWT.

10 Marks (CO1, CO2)

OR

- (b) A wind turbine has following data :

Speed of free wind at a height of 10 m = 12 m/s;

Air density = 1.226 kg/m^3 ; $\alpha = 0.14$; Height of tower = 100 m;

Diameter of rotor = 80 m; Wind velocity at the turbine reduces by 20%;

Generator efficiency = 85%.

Find :

- (i) Total power available in wind

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(2)

- (ii) Power extracted by the turbine
- (iii) Electrical power generated.

10 Marks (CO1, CO2)

3. (a) Draw and discuss the Power versus Wind-Speed characteristics of wind turbine clearly depicting the four separate regions of operation.

10 Marks (CO2)

OR

- (b) A wind turbine with three blades, each 4 m long, what distance does the tip of each blade travel in one full revolution ? If the turbine is rotating at the rate of 42 revolution per minute. How long does it take for one full revolution ? What is the speed of wind turbine if the wind is blowing at 6 m/s and also calculate the TSR of the turbine.

10 Marks (CO2)

4. (a) Discuss Betz criterion. What are the most favorable sites for installing wind turbines ?

10 Marks (CO3)

OR

- (b) A HAWT is installed at a location having free wind velocity of 15 m/s. The 80 m diameter rotor has three blades attached to hub. Find the rotational speed of the turbine for optimal energy extraction.

10 Marks (CO3)

5. (a) What are the deciding factors to be considered for the selection of generator in WECS ?

10 Marks (CO2, CO3)

OR

- (b) Discuss Scherbius Variable speed drive scheme. 10 Marks (CO2, CO3)

17/05/22

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TOE-802

B. TECH. (EE) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April/May, 2022

INTELLIGENT INSTRUMENTATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What is a process and a process parameter ? Give *two* examples of process parameters in each of the following processes : 10 Marks (CO1)
- (i) An automobile
 - (ii) An electric motor

OR

- (b) Classify the following process parameters under the two categories commonly used and rarely used :
- flow, pressure, texture, level, RPM, pH, and oxygen content.

10 Marks (CO1)

2. (a) What is the basic difference between a sensor and a transducer ? Give examples. 10 Marks (CO1)

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OR

(b) What do you understand by MEMS sensors ? Explain the working principle of MEMS sensors. 10 Marks (CO1)

3. (a) What is the basic difference between an analog and a digital transducer ? How can an analog transducer be converted to a digital one ? 10 Marks (CO1)

OR

(b) Explain the working principles of a semiconductor thermal sensor and a Hall effect sensor. 10 Marks (CO1)

4. (a) Explain the working of a smart pressure sensor with temperature compensation through proper block diagram. 10 Marks (CO2)

OR

(b) Distinguish between classical sensors and smart sensors. 10 Marks (CO2)

5. (a) What do you understand by biosensors ? How are they different from neuro-sensors ? 10 Marks (CO2)

OR

(b) List the various advantages of biosensors and neuro-sensors over classical sensors. 10 Marks (CO2)